

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution												
Correctly completes precedence table	AO1.1b	B1	<i>D</i> <i>C, E</i> <i>G</i> <i>F, H</i>												
Correctly states earliest start times for all activities	AO1.1b	B1	<table border="1"><tr><td>A</td><td>B</td><td>C</td></tr><tr><td>0 1 3</td><td>1 18 21</td><td>19 1 22</td></tr></table>	A	B	C	0 1 3	1 18 21	19 1 22						
A	B	C													
0 1 3	1 18 21	19 1 22													
Correctly states latest finish times for all activities	AO1.1b	B1	<table border="1"><tr><td>D</td><td>E</td><td>F</td></tr><tr><td>0 10 10</td><td>10 12 22</td><td>22 1 23</td></tr></table> <table border="1"><tr><td>G</td><td>H</td><td>J</td></tr><tr><td>0 4 11</td><td>4 12 23</td><td>23 2 25</td></tr></table>	D	E	F	0 10 10	10 12 22	22 1 23	G	H	J	0 4 11	4 12 23	23 2 25
D	E	F													
0 10 10	10 12 22	22 1 23													
G	H	J													
0 4 11	4 12 23	23 2 25													
States the correct activity (label or in words) from their completed activity network Only FT if their answer is A, D or G	AO1.1b	B1F	<i>D, Chop vegetables</i>												
Identifies their activity as critical OR States that the float of their activity is zero	AO2.5	E1F	<i>D is the first activity on the critical path</i>												
States their correct time using their completed activity network 6:30 pm + their minimum completion time Must see indication of pm OE	AO3.2a	B1F	6:55 pm												
Completes precedence table 2 with correct durations Condone correct durations in seconds	AO3.1b	B1	<table border="1"><tr><th>Duration</th><th>Immediate Predecessors</th></tr><tr><td>10.25</td><td>–</td></tr><tr><td>8.25</td><td><i>J</i></td></tr><tr><td>0.5</td><td><i>K</i></td></tr></table>	Duration	Immediate Predecessors	10.25	–	8.25	<i>J</i>	0.5	<i>K</i>				
Duration	Immediate Predecessors														
10.25	–														
8.25	<i>J</i>														
0.5	<i>K</i>														
Uses their minimum completion time and 19 in consistent units to find the float in the second project	AO1.1a	M1	$25 - 19 = 6$ minutes												
States the correct time CAO Condone omission of pm	AO1.1b	A1	6:36 pm												
Total 9 marks															

Q2.

Exam Papers Practice				
	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds the correct earliest start time for each activity	AO1.1b	B1	<pre>graph LR; A[A: 0 5 9] --> B[B: 0 4 4]; A --> C[C: 0 2 8]; A --> D[D: 5 3 12]; B --> E[E: 4 1 5]; C --> F[F: 2 8 10]; D --> G[G: 8 5 15]; E --> H[H: 8 6 18]; F --> I[I: 8 4 18]; G --> J[J: 14 2 17]; H --> K[K: 18 3 17]; I --> K</pre>
	Finds the correct latest finish time for each activity	AO1.1b	B1	
(b)	Evaluates the effects of reducing the duration of one of the activities on the project completion time	AO3.1b	M1	Reducing activity <i>E</i> duration to 1 hour reduces the project completion time to 14 hours, whereas all other activities reduce the project completion time to 15 hours or more
	Compares the effects of reducing the duration of each of the critical activities on the project completion time	AO3.1b	M1	
	Correctly deduces the activity which should have its duration reduced to one hour, from correct reasoning	AO2.2a	R1	
(c)	Correctly identifies a limitation in the context of the project	AO3.5b	B1	Time between one activity ending and the next activity starting is not taken into account, as workers may need to travel to a different location
	Explains how the limitation that has been identified affects project in time or monetary terms	AO2.4	E1	Not taking into account time. The travelling time will cause subsequent activities to be delayed, increasing the project completion time
Total 7 marks				

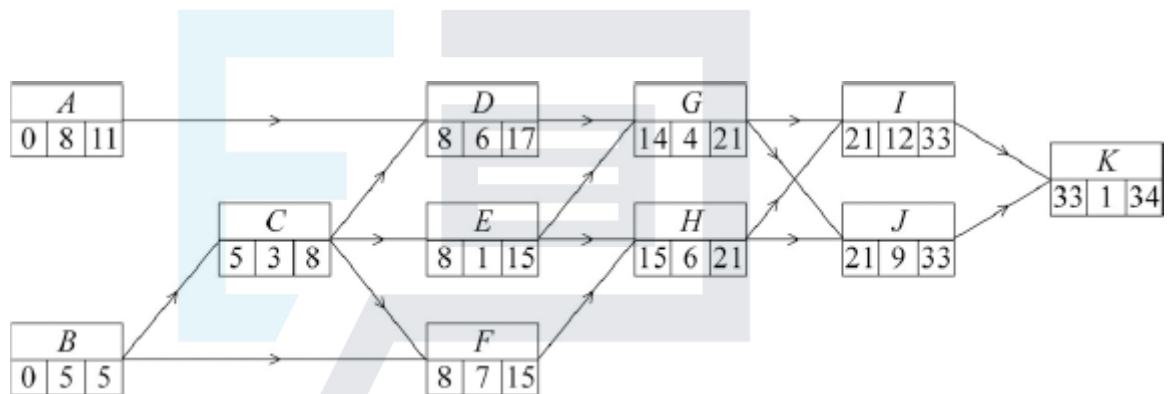
Q3.

Marking Instructions	AO	Marks	Typical Solution
Translates problem into that of finding a minimum spanning tree by listing or drawing 4 labelled arcs	AO3.1b	M1	

(Condone A for Alvanley etc)			
Finds correctly 4 arcs of the minimum spanning tree by listing or drawing	AO1.1b	A1	Alvanley to Helsby: 750 Elton to Ince: 1250 Alvanley to Dunham: 2000 Dunham to Elton: 2500
Determines correctly the total minimum length of cable required, complete with units	AO1.1b	B1	$750 + 1250 + 2000 + 2500 = 6500 \text{ m}$
Total 3 marks			

Q4.

(a)



Forward pass, correct at two of D, E, F

All correct

Backward pass, correct at G AND H ft

All correct

M1

A1

M1

A1

4

(b)

Activity	Predecessor
A	—
B	—
C	B
D	A, C
E	C

<i>F</i>	<i>B, C</i>
<i>G</i>	<i>D, E</i>
<i>H</i>	<i>E, F</i>
<i>I</i>	<i>G, H</i>
<i>J</i>	<i>G, H</i>
<i>K</i>	<i>I, J</i>

6+ correct

All correct

B1

B1

2

(c) (Critical) *B C F H I K*

B1

1

(d) (Float *E*) 6 (hrs)

B1

1

(e)



Their critical activities and 3 others shown

M1

Critical activities and 3 others correct

A1

All correct, condone floats seen

A1

3

(f) 34 (hrs)

B1

1

(g) 62 (hrs)

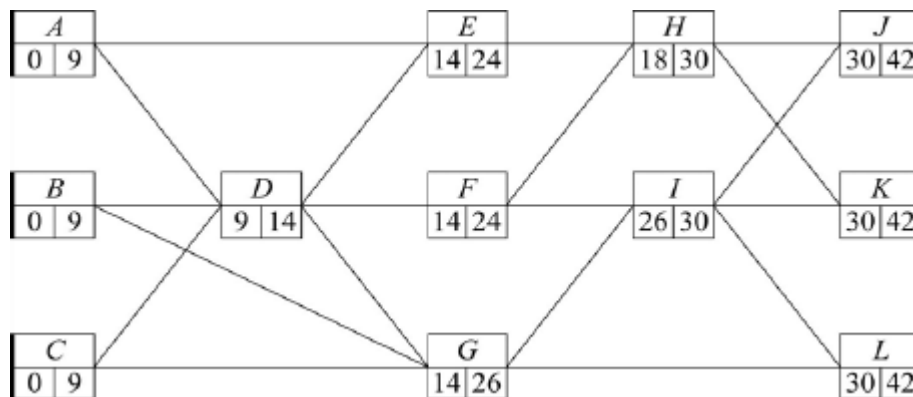
B1

1

[13]

Q5.

(a)



Forward pass, correct at D, E, F, G

M1

All correct

A1

Backward pass, correct at H, I, G ft

M1

All correct

A1

(b) C D G I J only

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B1

1

(c) 6

Their (latest – earliest – 4)

B1ft

1

(d) H delayed by 4

E1

K delayed by 5

B1

New time 51

51 scores 3/3

B1

3

[9]

Q6.

(a) $x = 4$

B1

$y = 12$

B1

$z = 13$

B1

3

(b) $B D H J$ and $C E I J$

first correct path

M1

2nd correct and no others

A1

2

(c) G

B1

Float = 3

B1

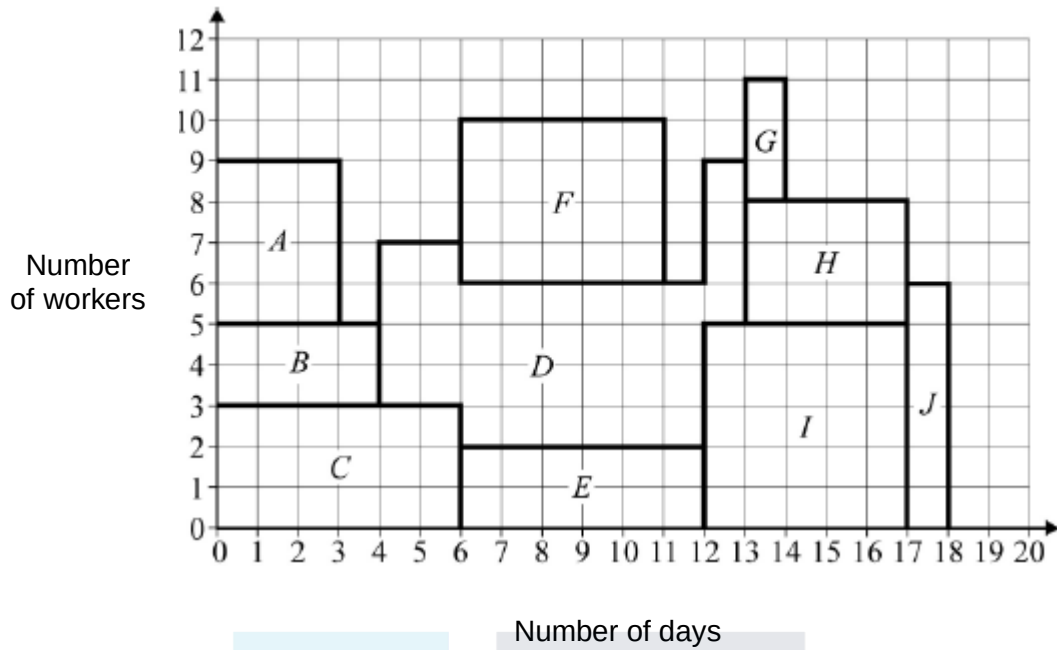
2

(d) One of their CPs correct height
and correct durations

M1

B, D, H, J and C, E, I correct
and correct durations

A1



A starting at 0 and ending at 3

one correct with correct height

M1

F starting at 6 and ending at 11

two correct with correct height

A1

G starting at 13 and ending at 14

all correct with correct height

withhold first A1 earned if it is not clear which activities take place at any given time

withhold another A1 if "holes" appear in histogram

A1

5

(e) New earliest J 22 days

assuming activities continuous

B1

Minimum extra time 5 days

assuming activities continuous

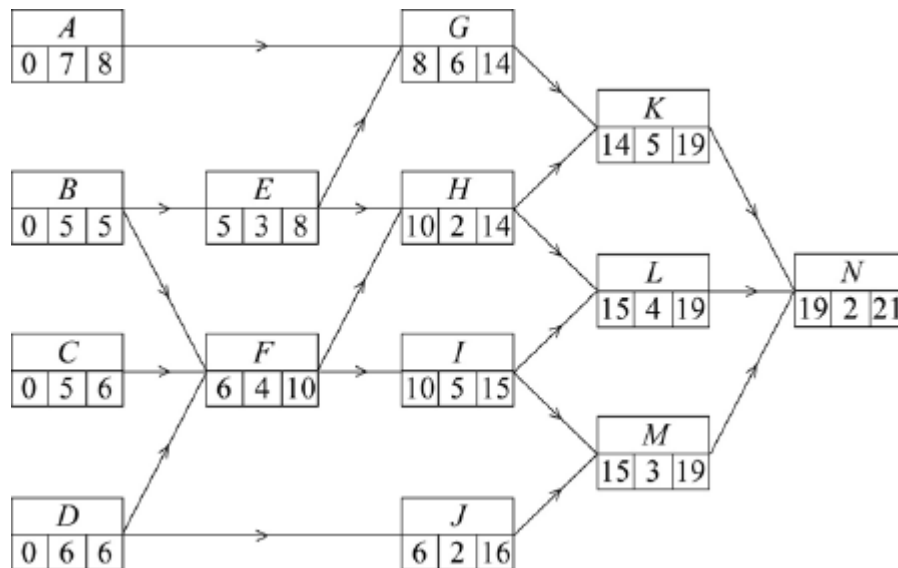
B1

2

[14]

Q7.

(a)



Forward pass

condone one slip (follow through)

all correct

M1

A1

Backward pass

condone one slip (follow through)

all correct

M1

A1

4

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(b) Critical paths

B E G K N

first path correct

M1

D F I L N

second path and no others

A1

Minimum completion time is 21 days

B1

3

(c) Cascade diagram

may be in blocks or bars (see examples)

One of 'their' CPs correct
ft their CP

M1

B , D , E , F , G , I , K , L , N
these activities correct

A1

A , C , H , J , M
3 of these with correct start and duration

M1

3 correct with correct slack indicated

A1

all 5 correct with correct slack

A1

		Slack
A	0 – 7	7 – 8
C	0 – 5	5 – 6
H	10 – 12	12 – 14
J	6 – 8	8 – 16
M	15 – 18	18 – 19

5

(d) (Max value of x is) 10

considering $J_{latest} - J_{earliest}$

M1

$\Rightarrow x \leq 10$

(condone $x < 11$ for SC2)

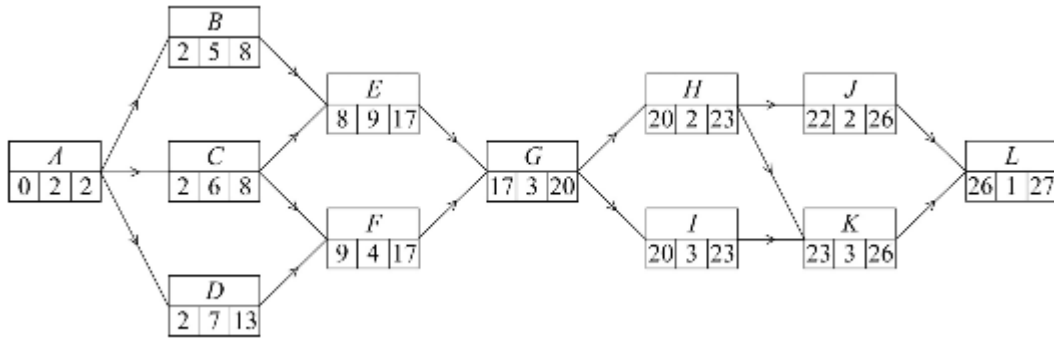
NMS $x \leq 10$ award M1 A1

A1 cao

2

[14]

Q8.



- (a) Forward pass
up to one slip ft

M1

Correct

A1

- Backward pass
up to one slip ft

M1

Correct

A1

4

- (b) (i) Critical path $A C E G I K L$

B1

- (ii) Float for $D = 13 - 2 - 7$

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M1

= 4 days

A1

3

- (c) $A C E G I K L$ correct durations
and heights
one slip in duration or height

M1

correct

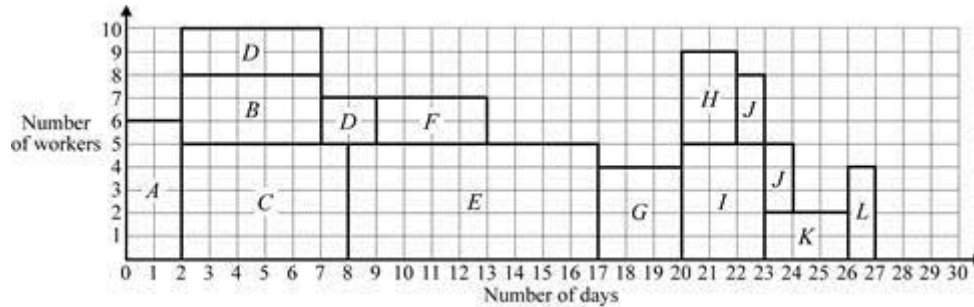
A1

D and B and F correct (no "holes")

B1

*H and J correct (no "holes")
withhold final mark earned if not clear which
activities are taking place at any time*

B1



4

- (d) Correctly dealing with *D*, *B* and *F*
ft 1 slip

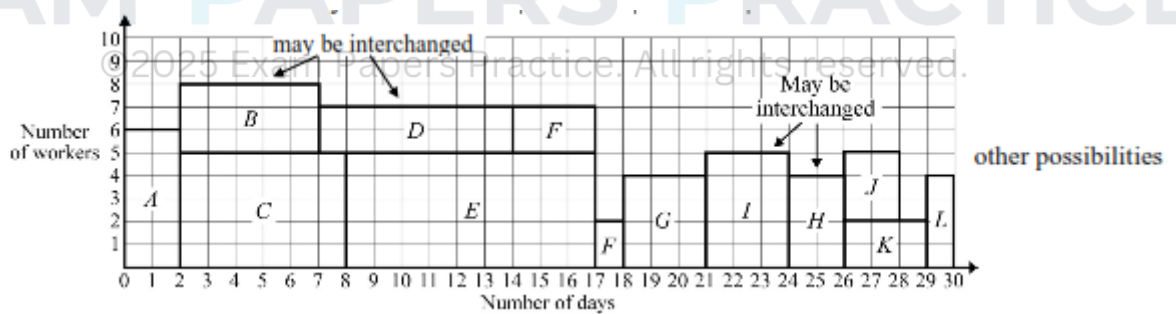
B1

Correctly dealing with *H* and *J*
ft 1 slip

B1

Minimum extra time = 3 days
CAO

B1

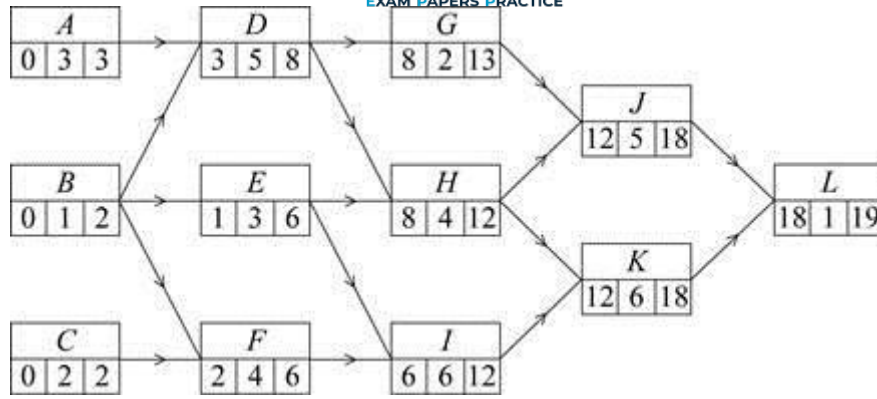


3

[14]

Q9.

(a)



Earliest start times

condone one slip + ft

M1

all correct

A1

Latest finish times

condone one slip + ft

M1

all correct

A1

4

(b) Critical paths *ADHKL*

one path correct

B1

CFIKL

second path correct and no others

B1

Minimum time = 19

19 days

B1

3

(c) Greatest float time at G

ft their activity with greatest float for M1

M1

$(13 - 8 - 2) = 3$ (days)

values at G must be correct

A1cso

2

(d) one of 'their' critical paths "correct"

M1

A, D, H, K, L and C, F, I

all 8 of these activities correct

A1

B, E, G, J

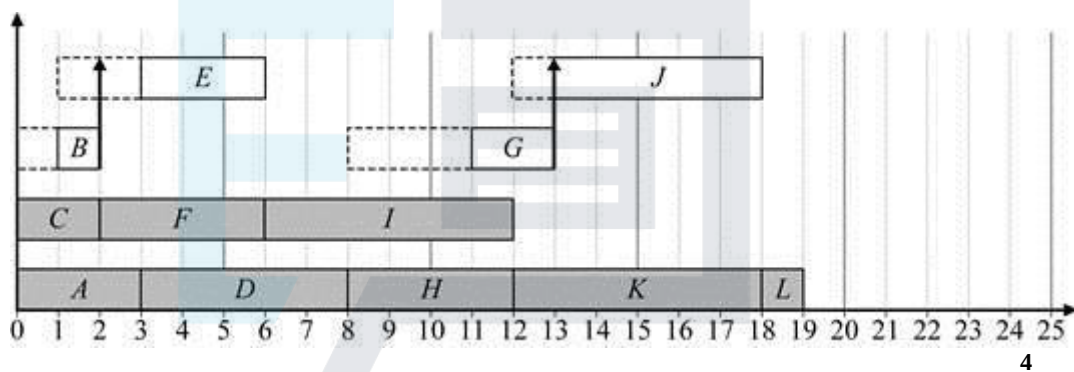
B(1-2); E(3-6); G(11-13); J(13-18)

3 of these with correct duration and latest start time (may omit slack)

M1

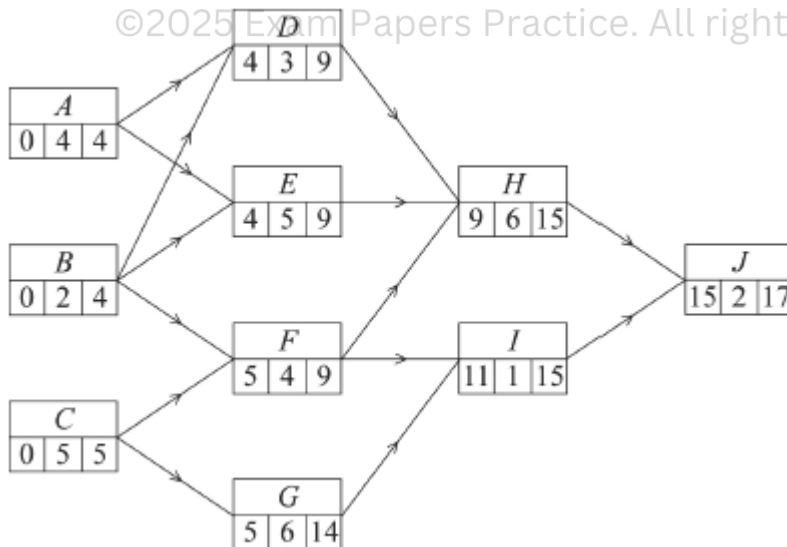
all 4 correct with correct slack shown

A1cso



[13]

Q10.



(a) (i) Earliest start times

At least 4 correct (other than zeros)

M1

All correct

A1

2

(ii) Latest finish times

Up to one error + follow through

M1

All correct

A1

2

(b) Float for $G = 3$ days

ft their G values

B1ft

1

(c) Critical paths $A E H J$

and $C F H J$

Minimum completion time = 17 days

First correct path

B1

Second path (and no others)

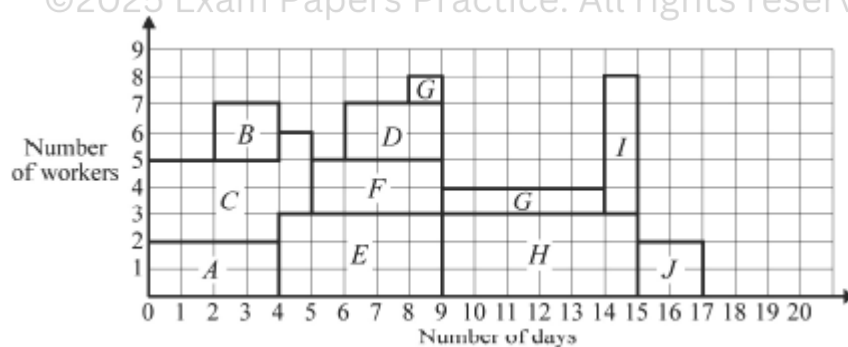
B1

Must be stated explicitly

B1

3

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One of their CPs – correct heights

NB “holes” penalise first A1 earned

M1

$A E H J$ and $C F (H J)$ correct

A1

B starting at 2 (and ending at 4)

D starting at 6 (and ending at 9)
G starting at 8 (and ending at 14)
I starting at 14 (and ending at 15)
One correct with correct height

M1

Two correct with correct height

A1

All correct with correct height and no slack
Withhold first A1 earned if it is not clear
which activities take place at any given time

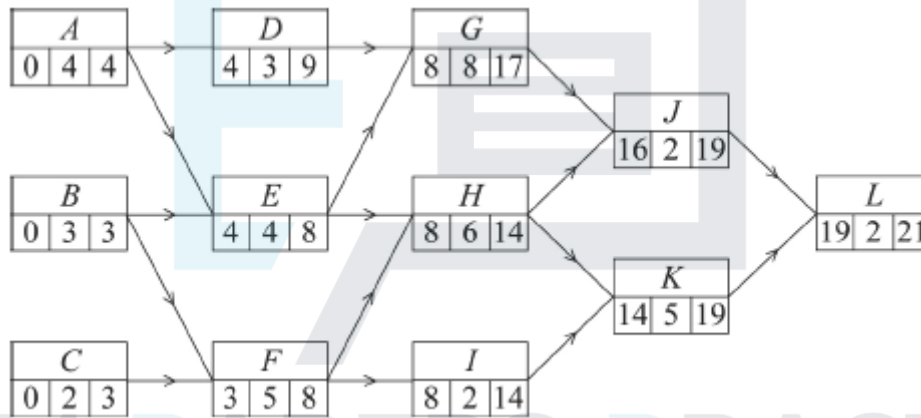
A1

5

[13]

Q11.

(a)



Earliest start times

one slip follow through

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M1

all correct

A1

Latest finish times

one slip follow through

M1

all correct

A1

4

(b) Critical paths are *AEHKL* and *BFHKL*

one correct

M1

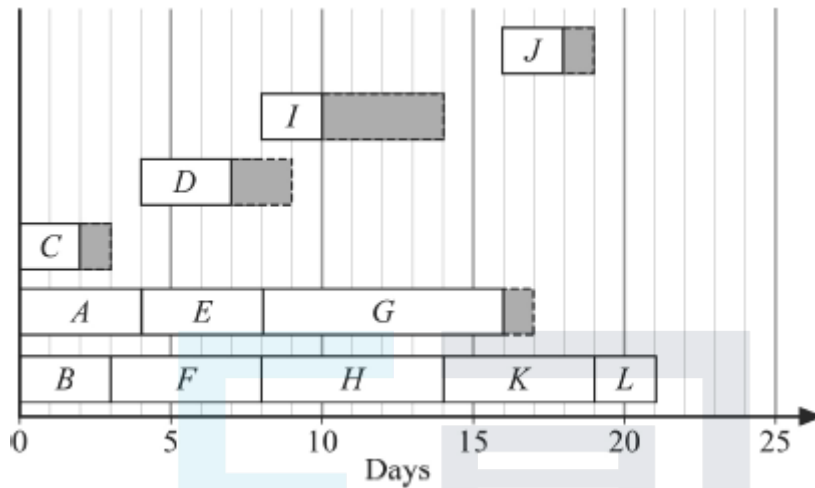
A1

Minimum completion time = 21 days

B1

3

(c)



A (0 → 4)

B (0 → 3)

C (0 → 2 → 3)

D (4 → 7 → 9)

E (4 → 8)

F (3 → 8)

G (8 → 16 → 17)

H (8 → 14)

I (8 → 10 → 14)

J (16 → 18 → 19)

K (14 → 19)

L (19 → 21)

A, B, E, F, H, K, L correct

B1

C, D, G, I, J

(4 with correct start and duration)

M1

All 5 correct with correct slack indicated

A1

3

(d) (i) K now starts day 17

or "delayed" b 3 days if 14 in network

B1

L now starts day 22
or "delayed" b 3 days if 19 in network

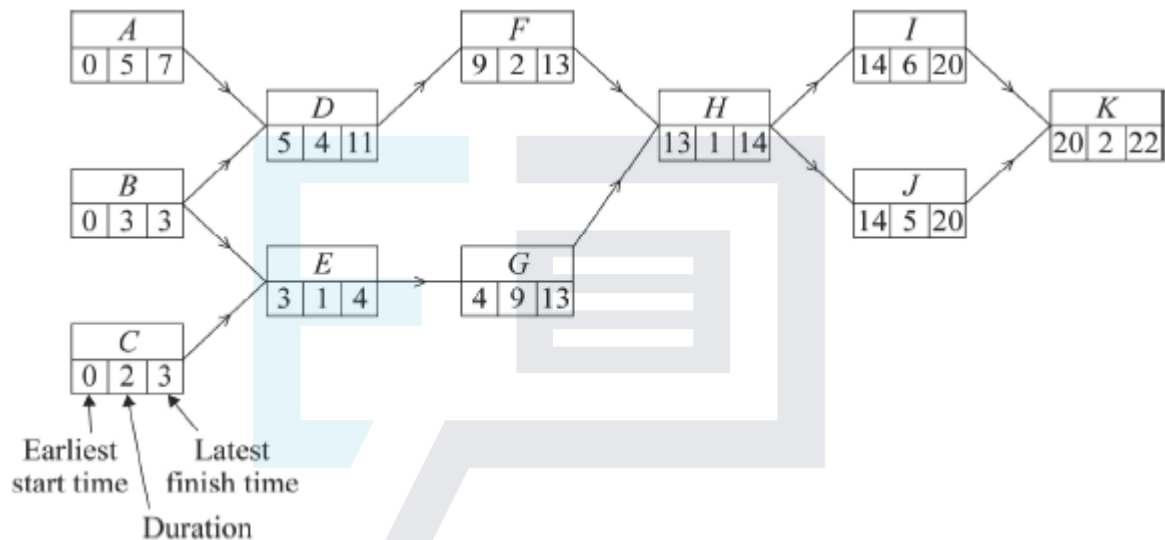
B1
2

(ii) Overall delay 3 days

B1
1

[13]

Q12.



(a) Network attempted (3 more activities)

SCA

M1

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Up to 2 slips (boxes or connections)

A1

Correct network

Condone missing arrows if sequence is clear

A1
3

(b) (i) Forward pass
up to 1 slip ft

M1

Correct

A1
2

- (ii) Backward pass
up to 1 slip ft

M1

Correct

A1

2

- (c) Minimum completion time 22 days
Must be stated – not simply in K box

B1

Critical path *B E G H I K*
and no others

B1

2

- (d) (i) New start time for *H* is 15 days
For *H*, their (*F* earliest time 9) + (2 + 4)

M1

New start time for *I* is 16 days
both correct

A1

2

- (ii) Minimum delay is 2 days
Condone new completion time 24 days

B1

1

[12]

Q13.

- (a) *G*, *H* and *I* in correct place

M1

Lines (with arrows) correct

A1

2

- (b) Forward pass (no more than 1 error FT)

M1

Early start times correct

A1

Backward pass (no more than 1 error FT)

M1

Latest finish times correct

A1

See below

4

(c) Correct critical path: *ACEGI*

B1

Correct minimum time: 25 days

B1

2

(d) "Their" critical activities

See below

B1ft

Block $0 \leq t \leq 10$

B1

$10 \leq t \leq 11$

B1

All correct including labels

CSO

B1

4

(e) Problem with *F* or day 11

M1

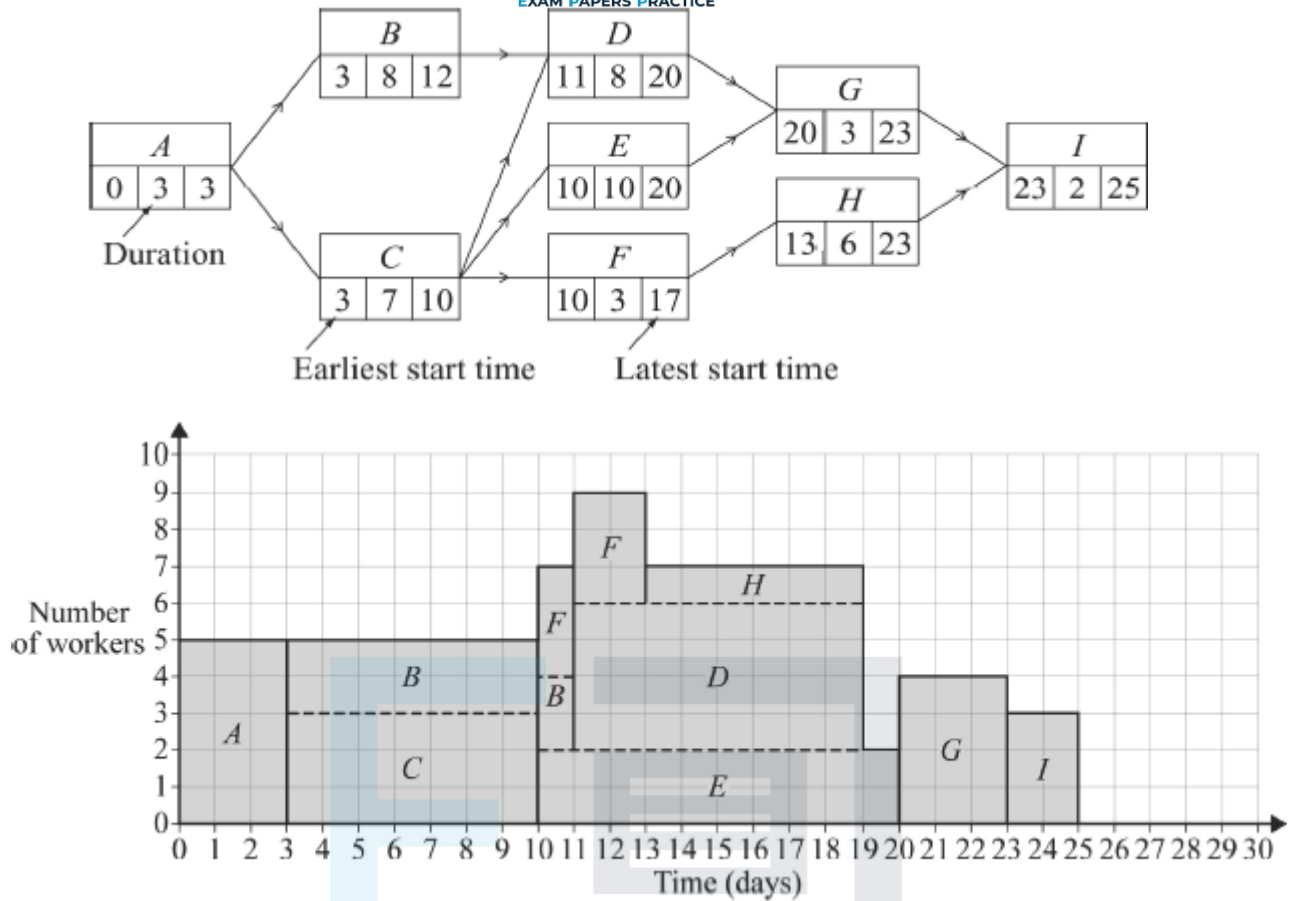
Delay start of *D* (by 2 days), then *G* and *I* (by 1 day)

A1

Extra time 1 day

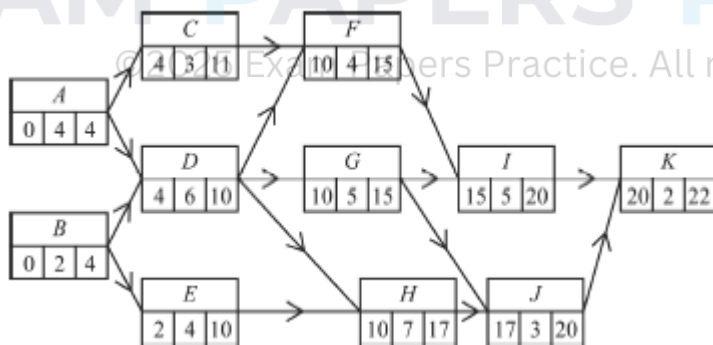
B1

3



[15]

Q14.



- (a) Earliest start times
up to 2 errors ft

M1

All correct

A1

Latest finish times
up to 2 errors ft from K

M1

Correct

A1

4

- (b) Critical paths $A D G I K$
 $A D H J K$
withhold if extra path such as
 $A D G J K$ given

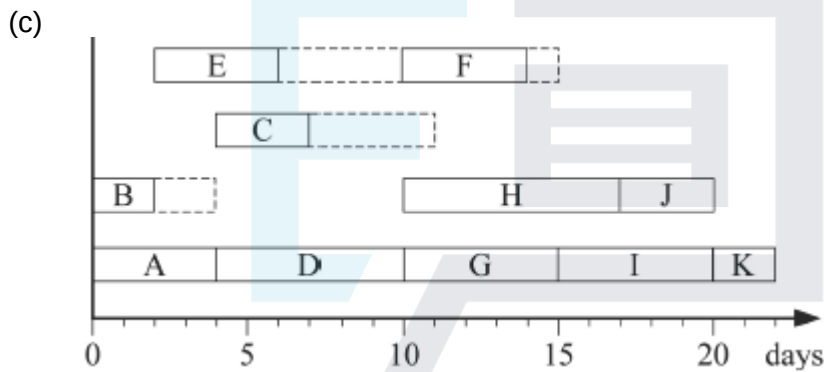
B1B1

Minimum time for completion = 22 days

Must be stated – not just a value in box

B1

3



A, D, G, H, I, J, K correct

B1

B, C, E, F Exam Papers Practice. All rights reserved.

2 correct with slack /float or 4 correct & no slack

B1

all correct with slack/float; withhold if slack not shown dotted etc

B1

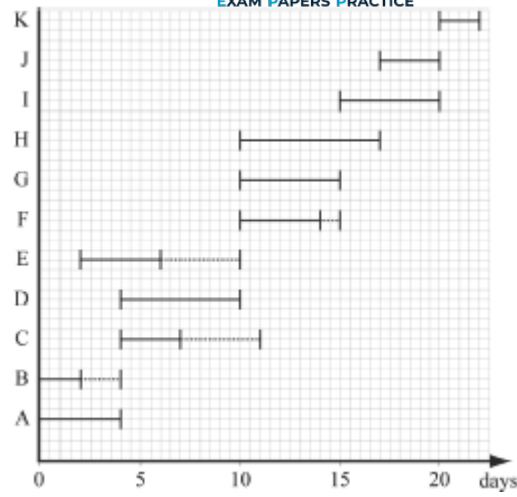
3

- (d) F starts after 12 days at earliest
 or F starts 2 days later
 I is now unable to start until after 16 days
 or starts 1 day later

E1

Minimum completion time now 23 days – one extra day etc

B1



2

[12]

Q15.

(a) Network attempted
SCA

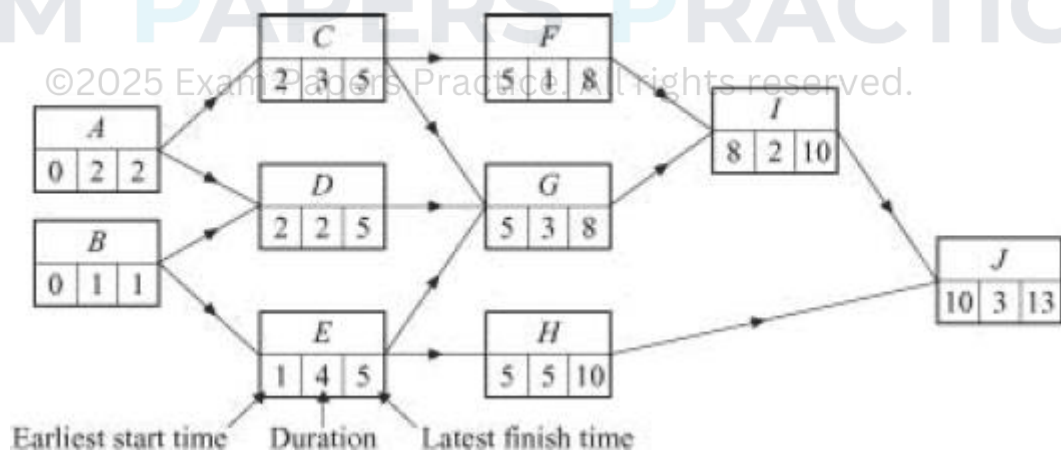
M1

up to 2 slips (boxes or arrows)

A1

correct network

A1



3

(b) Forward pass

M1

correct

A1

2

(c) Backward pass

M1

correct

A1

2

(d) Minimum completion time: 13 weeks

B1

Critical paths: *ACGIJ*

B1

BEGIJ

B1

BEHJ

B1

4

[11]

Q16.

(a)

Activity	Immediate Predecessors
<i>A</i>	–
<i>B</i>	–
<i>C</i>	<i>A, B</i>
<i>D</i>	<i>B</i>
<i>E</i>	<i>B</i>
<i>F</i>	<i>C</i>
<i>G</i>	<i>D</i>
<i>H</i>	<i>D, E</i>
<i>I</i>	<i>F, G</i>
<i>J</i>	<i>G, H</i>
<i>K</i>	<i>I, J</i>

Up to 2 slips

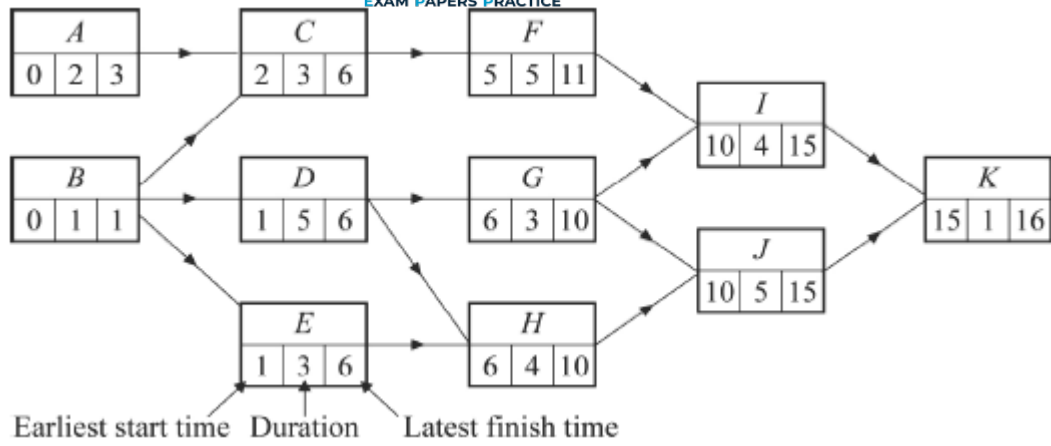
M1

All correct

A1

2

(b)



Start times – up to 1 slip with FT

M1

All correct

A1

Finish times – up to 1 slip; FT 'their 16'

M1

All correct; CSO

A1

4

(c) Critical path *B D H J K*

B1

Minimum time 16 days

B1

2

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(d) Greatest float at *E*

B1ft

Value = 2

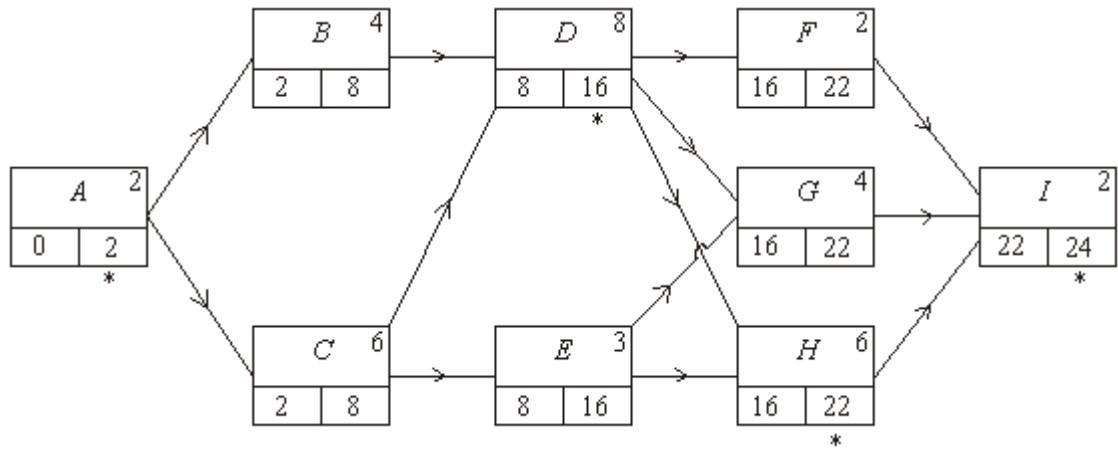
B1ft

2

[10]

Q17.

(a)

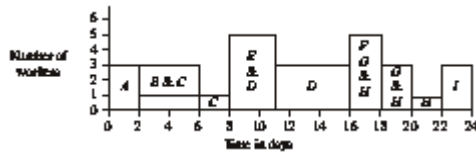


Activity network SCA

- (a) Forward pass for earliest times
- almost correct (up to 2 slips) M1
- all correct A1
- (b) Forward pass for earliest times A1 3
- (c) Backward pass M1
- all correct A1 2
- (d) Critical path is ACDHI B1
- Minimum completion 24 days B1 2
- (e) Non-critical Float B E F G
- Float 2 5 4 2

At least 3 activities and float in one activity ft correct

ft their earliest and latest times



(f)

Histogram ≤ 11

M1

Correct

A1

Rest as histogram – generally start activities ok

M1

All correct

A1

4

Resource histogram

(g) Problems with D & E solved by E coming after D

M1

Problem at 16-18 days with F can be solved by moving F to 20-22

A1

Must overrun by equivalent to duration of E (3 days)

B1

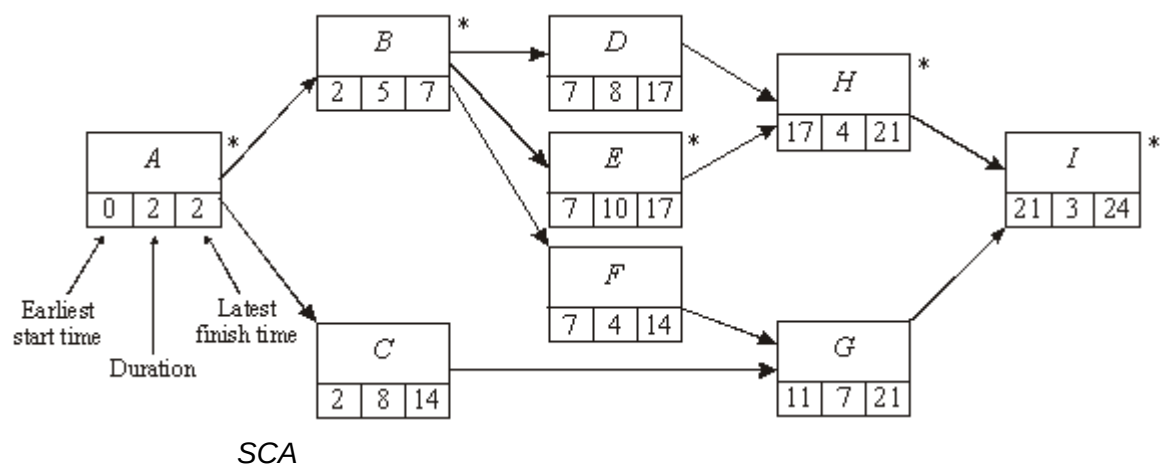
3

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[18]

Q18.

(a)



M1

(almost correct 2 slips)

A1

Correct

A1

3

(b) Forward pass for earliest start times

M1

All correct

A1

2

(c) Backward pass for latest finish times

M1

All correct

A1

2

(d) Critical path *A B E H I*

B1

1

(e)

Non critical *C D F G*

Float 4 2 3 3

At least one float time correct

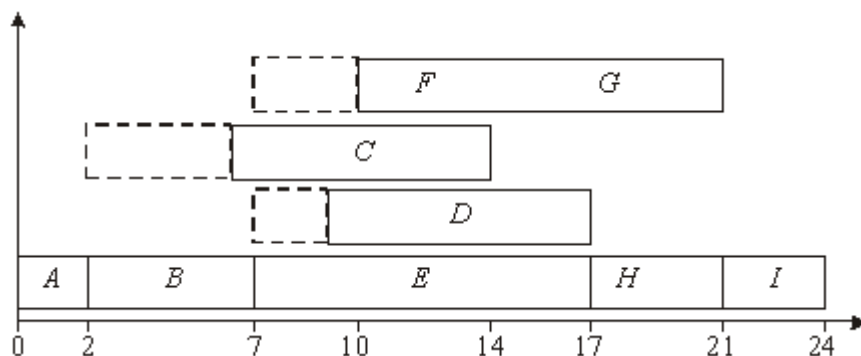
M1

All correct

A1

2

(f)



‘their’ critical path on chart

B1ft

C from 6 to 14 (with space 2-6)

One other activity (condone no slack or earliest start)

M1

D from 9 to 17 (with slack 7-9)

2 other non critical activities

A1

F & G from 10 to 21 with appropriate slack

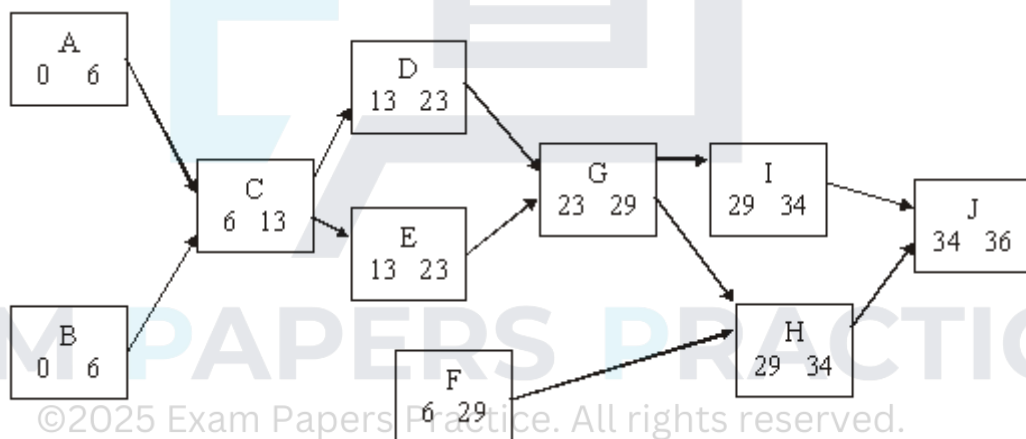
All correct

A1

4

[14]

Q19.



(a) Network

SCA

M1

–1 ee

A3

4

(b) Forward pass

M1

All correct

A1

2

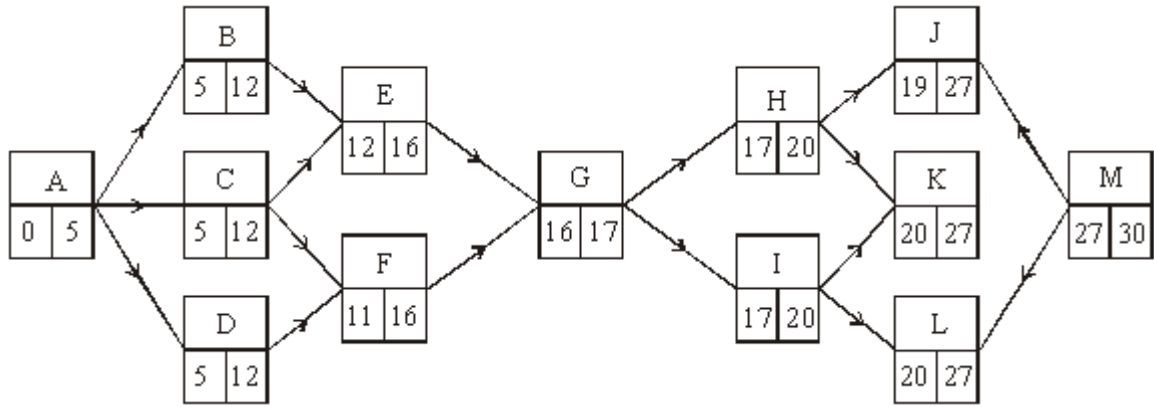
(c) Backward pass

				M1	
		<i>All correct</i>			
				A1	
					2
(d)	(i)	Project completion time	36 hours		
				B1ft	
					1
	(ii)	Critical path	<i>BCEGHJ</i>		
		SCA			
		<i>All correct</i>		M1	
				A1	
		Earliest start + activity duration = latest finish time		E1	
					3
(e)	(i)	<i>I now has new earliest time 29 + 3</i>			
		<i>Extra 3 hours on edge HI or new activity between H and I of duration 3</i>			
				M1	
		= 32			
				A1	
					2
	(ii)	<i>I now becomes critical and increases J earliest start time to 35</i>			
				M1	
		New completion time is 37 hours			
				A1	
					2

[16]

Q20.

(a)



Forward pass

Back pass

M1 A1

2

M1 A1

2

(b) Critical path *ABEGILM*

B1

1

(c) *D*

B1

1

(d) (*C* now 9, so *E* starts at 14)

SC 32 scores ½

M1

Overrun 2 days

A1

2

(e) *F* start at 14

SC 35 scores 2/3

M1A1

∴ overrun 5 days

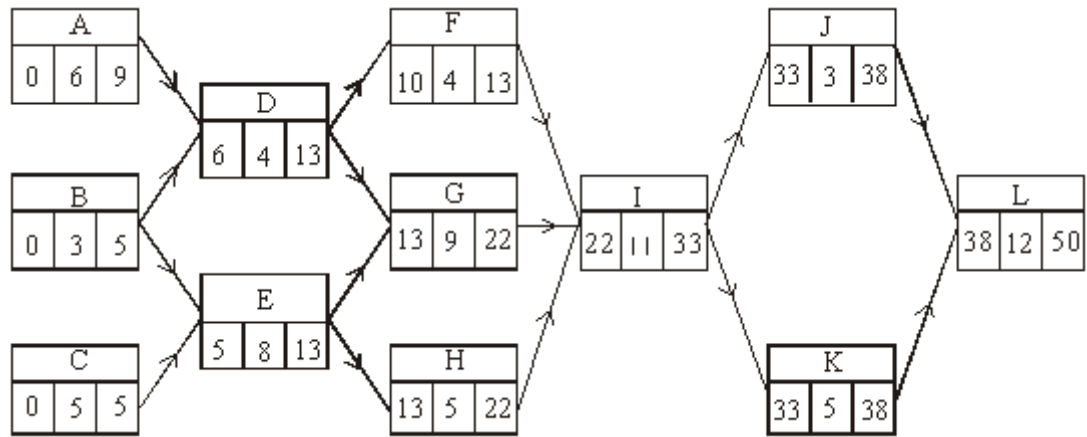
A1

3

[11]

Q21.

(a)



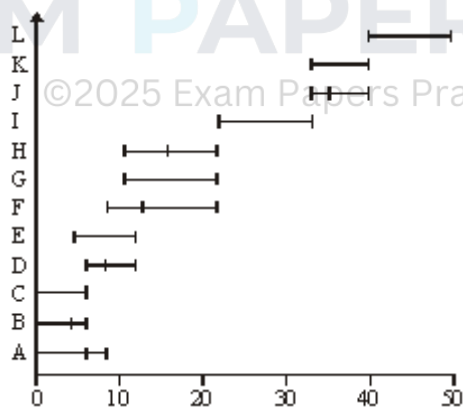
forward pass

back pass

(b) C E G I K L

(c) F

(d)



Gantt diagram

floats included

correct, excluding floats

M1A1

2

M1A1

2

B1

1

B1

1

M1

B1

A1

3

(e) (i) D 5 days $\Rightarrow G$ back 2 days

E1

$\therefore J$ starts at 35
OE

M1

$\therefore L$ starts at 43

$\therefore$ finish at 55

A1

3

(ii) $ADGIJL$

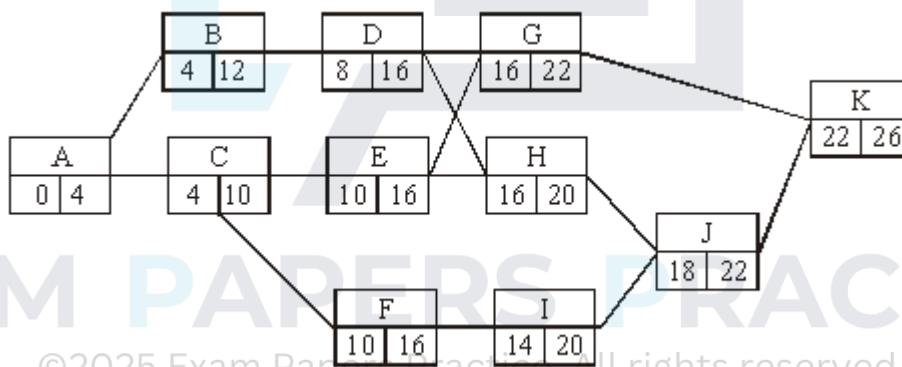
B1

1

[13]

Q22.

(a)



(i) *Forward*

M1A1

2

(ii) *Back*

M1A1

2

(b) CP = A C E G K
Min time = 26
Both

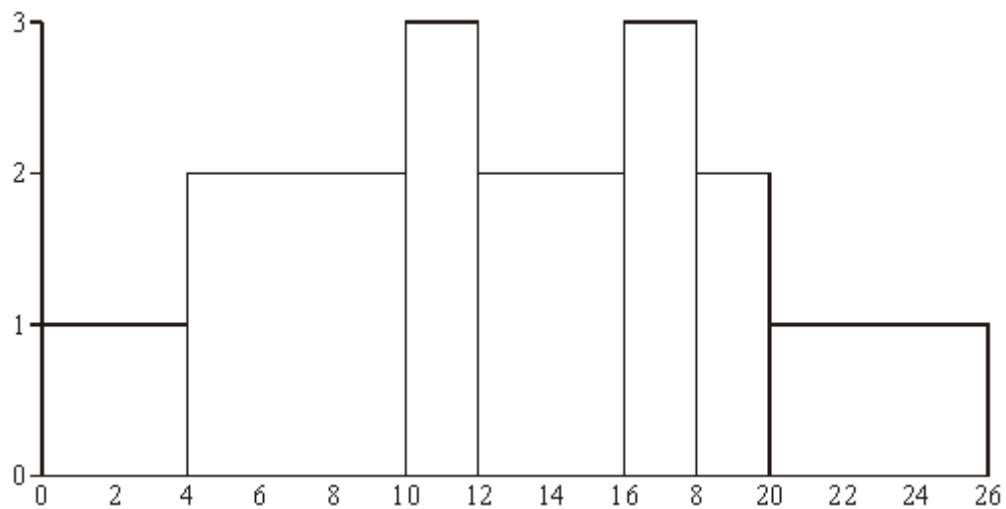
B1

1

(c) B, D

B1

(d)



-1 EE

M1A2

3

(e) (i) Min time = 28

B1

1 worker critical path

other activities $\Rightarrow$ J takes until 24

$\therefore$ overrun = 2

E1

2

(ii) A C E G K

and

B D F (H) I J

H either

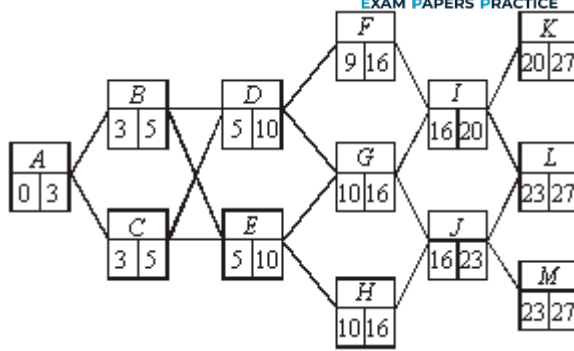
B1

1

[12]

Q23.

(a) (i) & (ii)



Forward

M1

A1

2

Back

M1

A1

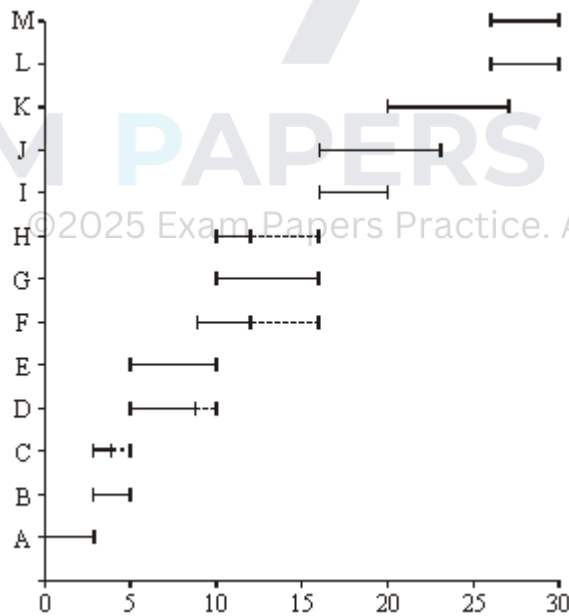
2

(b) C, H, F, D

B1

1

(c)



M1

(-1 EE)

A2

3

(d) (i) Extra at KLM

M1

Min extra 4
Total = 31

A1

2

(ii) A B D F H I K
 C E G J L M

B1

OE

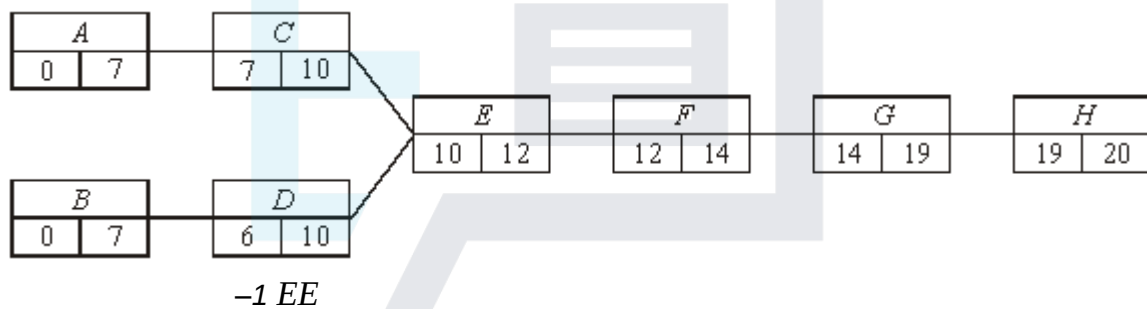
B1

2

[12]

Q24.

(a)



M1 A2

3

(b) Min = 20

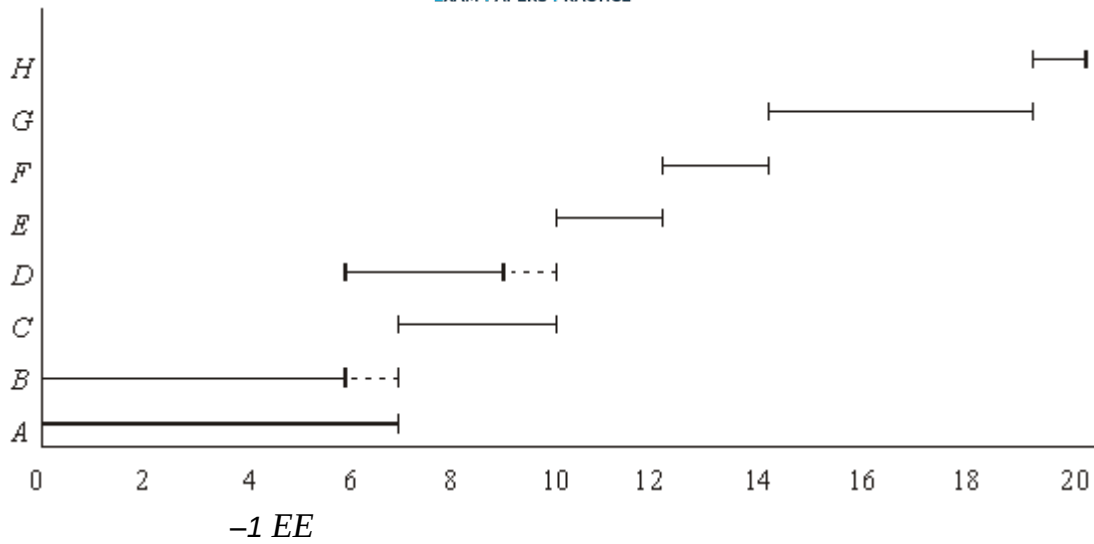
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M1 earned in (a)

M1 A1

2

(c)



M1 A3F

4

[9]

Q25.

(a)

Activity Immediate Predecessor

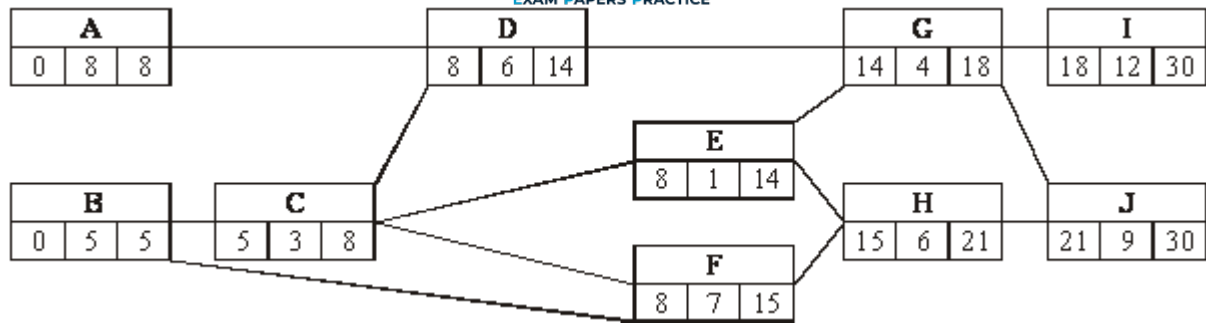
A	-
B	B
C	A, C
D	C
E	(B,) C
F	D, E
G	E, F
H	G
I	G, H
J	

-1 EE

B2, 1

2

(b)



Forward pass

Back pass

M1 A1

M1 A1

4

(c) Critical A, B, C, D, F, G, H, I, J

B1

1

(d)



-1 EE

M1 A3F

4

(e) Problem with D E F

M1

∴ overrun = 1 (hour)

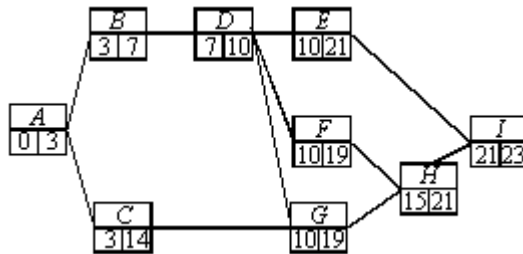
A1

2

[13]

Q26.

(a/b/c)



SCA

-1 EE

for forward pass

for all correct

for back pass

for all correct

M1

A2

3

M1

A1

2

M1

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A1

2

- (d) C – 9
F – 5
G – 4
H – 4

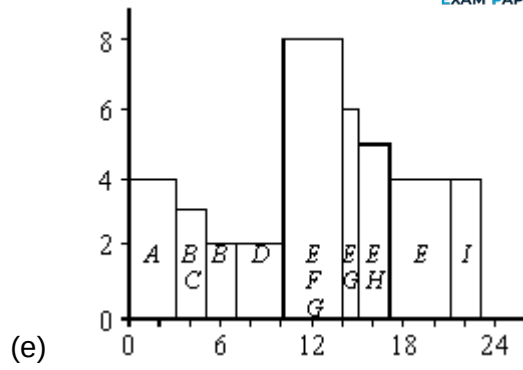
for all four letters

2

for four correct floats

B1F

B1



M1

SCA

M1

-1EE

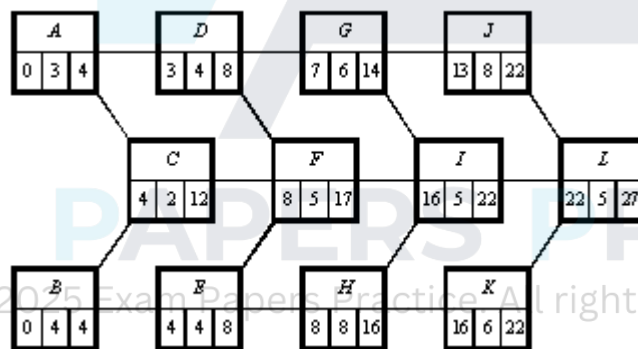
A3F

4

[13]

Q27.

(a) (i)



Start times

M1 A1

2

Finish time

M1 A1

2

(b) $B E H K L (= 27)$

B1

1

(c) (i) New min time = 24

M1

Path $B E H I L$

A1
2

(ii) Leave A, C, F (non crit)

B1

Speed B by 2 (crit)

B1

J by 1 (becomes crit)

B1

K by 1 (non crit)

Reason

B1B1

5

(iii) Cost = £79 000

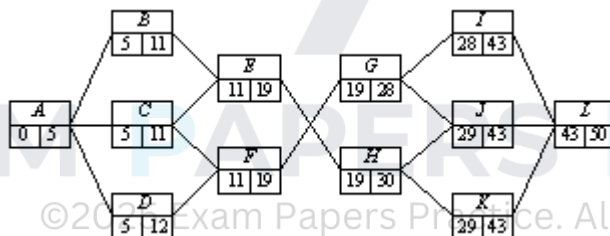
B1

1

[13]

Q28.

(a) (i)



start times

M1

A1

2

(ii) finish times

M1

A1

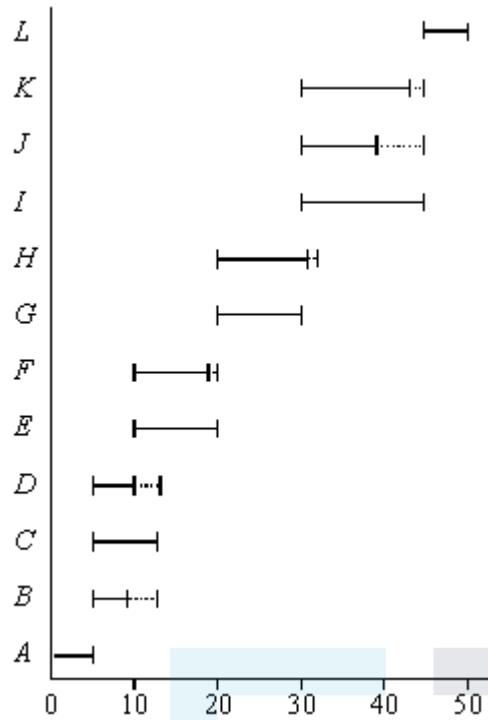
2

(b) Critical path A C E G I L

B1

1

(c)



- (d) From network
Problem with B, C, D and I, J, K

Minimum extra B, J

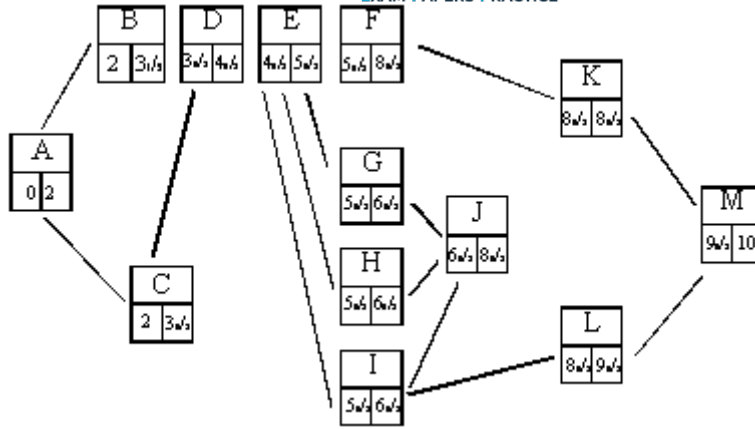
$$= 50 + 1 + 6 = 57$$

$$\begin{cases} B + D - C = 1 \\ J + K - I = 6 \end{cases}$$

Q29.

(a)

[11]



SCA

M1

– 1 each independent error

A3

4

(b) As diagram

– 1 EE
A2

M1 3

(c) As diagram

– 1 EE

M1

3

A2

(d) Critical A, C, D, E, H, I, J, K, M

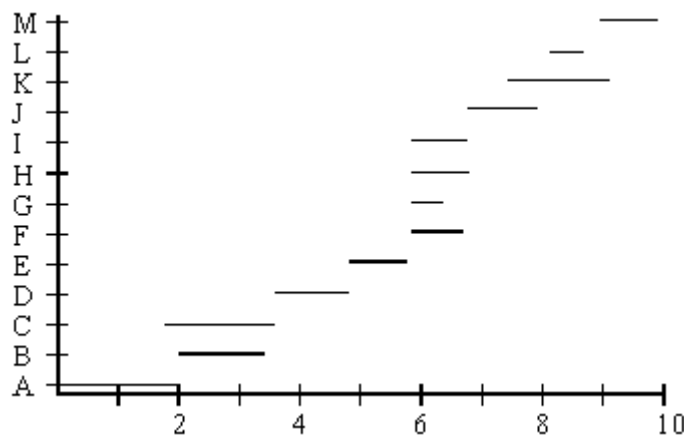
B1

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10 days

B1

2

(e)



SCA

-1 EE

M1

A2f

3

(f) Electrician B, G - $\frac{3}{4}$ day

M1

Plumber C, (I) - $\frac{3}{8}$ day

A1

Joiner (H), J - $\frac{2}{5}$ day

M1

Use joiner – electric not critical

A1

New completion time $9\frac{3}{5}$

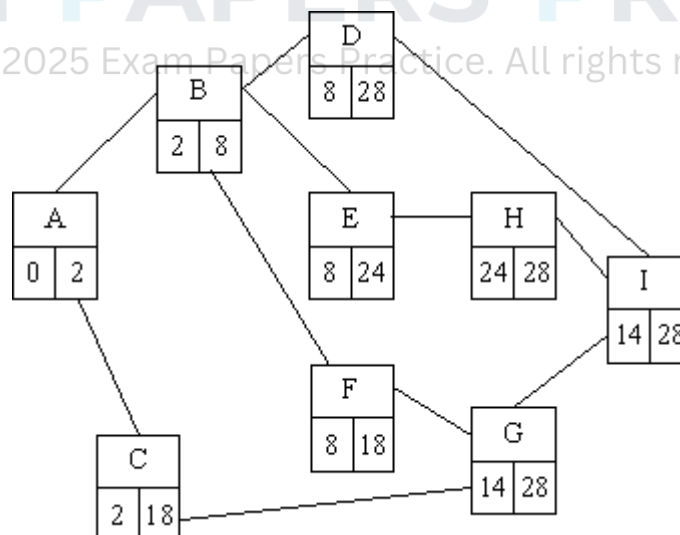
cao

5

[20]

Q30.

(a) (i)



-1 each independent error

M1 A3

4

(ii) As diagram

-1 EE

M1

A2

3

(iii) As diagram

M1

-1 EE

A2

3

(iv) A B E H I

Allow answers on diagram

B1

B1

2

(b)



SCA

M1

-1 EE

A3

4

(c) Must reduce 12 → 8 by 4 workers

B1

4

Can't move C, E not enough float

B1

Can move D back 4 days to 8
or F and G back 4 days
6 days 4 days

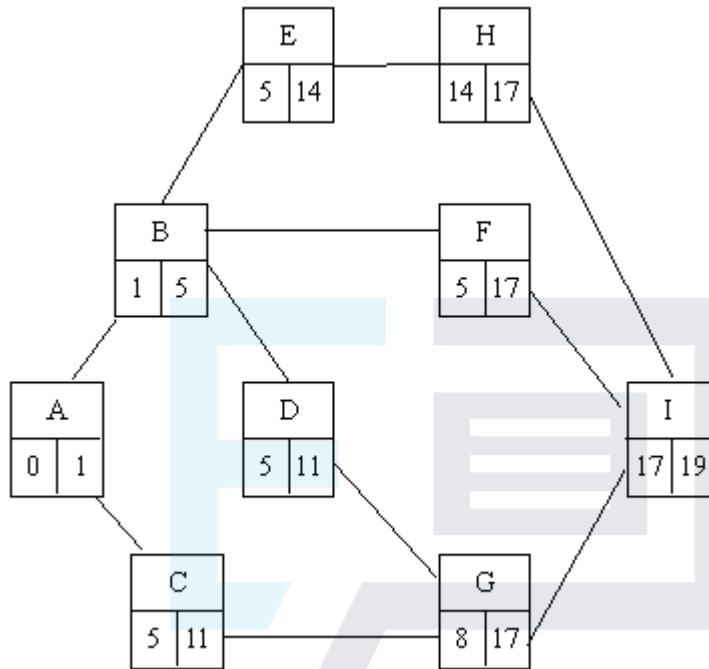
B1

B1

[20]

Q31.

(a)



– 1 each independent error

EXAM PAPERS PRACTICE

M1A3

4

(b) As diagram

M1A2

3

(c) As diagram

M1A2

3

(d) ABEHI

B1

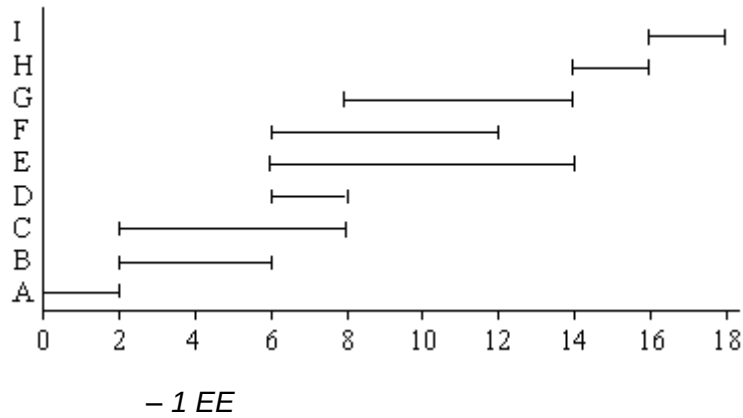
2

19 days

allow marked on diagram

B1

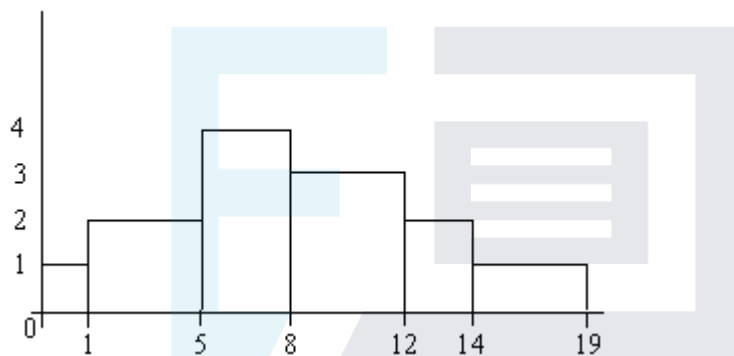
(e)



M1A3F

4

(f)



M1A1

2

(g) Worker 1 ABEHI

M1

2

Worker 2 CG

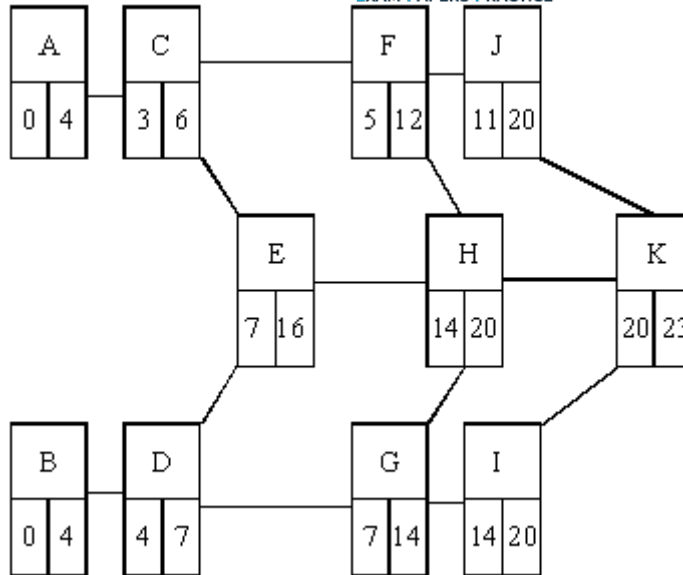
A1

Worker 3 DF

[20]

Q32.

(a)



-1 each independent error

(b) As diagram

M1A3

4

(c) As diagram

M1A2

3

(d) BDGIK

M1A1

3

23 weeks

B1

B1B2

2

(e) (i) Leave E, float time 4

E1

Hurry I by 2 on C.P.

E1

Hurry J by 1 more on C.P

E1

No more because of H

E1

4

Shortest time 21 weeks

B1

1

(ii) $4000 \times 2 + 3000$

M1

$= 11000$

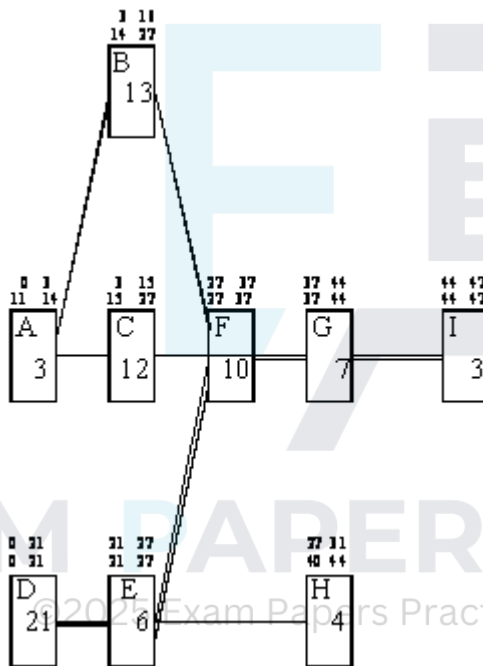
A1

2

[19]

Q33.

(a)



Attempt at a network with all 9 activities included,
starting at A and D, finishing at I

M1

All correct, – 1 EE

A3

4

(b) Upper figures show earliest times
Evidenced by labels

M1

(forward scan)

All correct

A1

Lower figures show latest times
Evidenced by labels

M1

(backward scan)
All correct

A1

Critical path *D-E-F-G-I*
Critical path stated and shown. ft c's labels provided upper and lower figures match

A1FA1F

6

(c)

A	B	C	D	E	F	G	H	I
11	11	12	0	0	0	0	13	0
	- 1	EE						

B2

2

(d)

Cascade diagram

Attempt a cascade diagram with all activities and a suitable time scale

M1

All correct. - 1 EE

A3

Minimum number of helpers: 3

No. of helpers (can be earned independently).

B1

5

(e) Least delay: 1 day
Delay

B1

Achieved by starting A and B as early as possible
 then do C immediately after B
OE involving B and C sequential rather than parallel

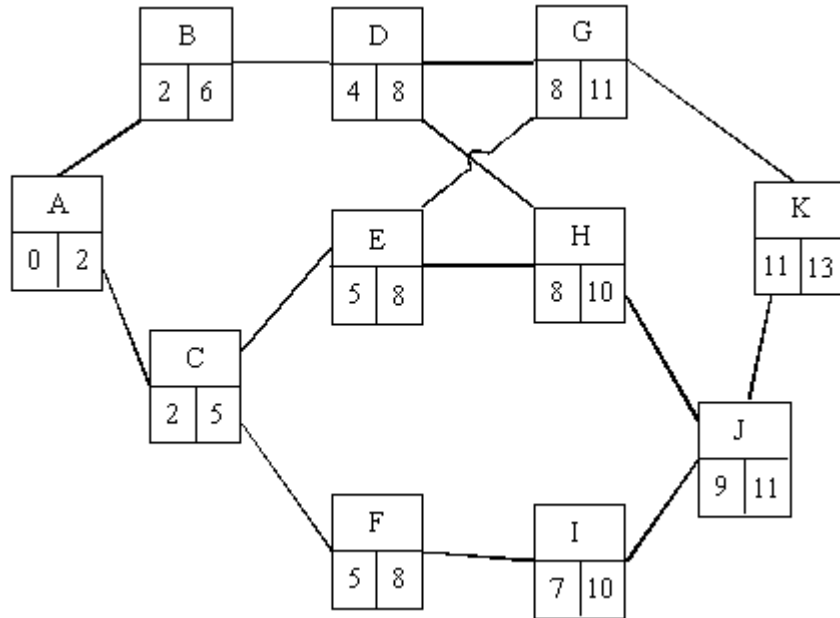
E1

2

[19]

Q34.

(a)



– 1 each independent error

(b) As diagram

A2 – 1 EE

M1A3

4

M1A2

3

(c) As diagram

A2 – 1 EE

A1A2

3

(d) ACEGK
13 days

B1

2

Allow marked on diagram

B1

(e) overrun B and D

B1B1

2

(f) (i) 14 days

B1

2

K held up 1 day

B1

(ii) ACEGK
BDFIJH

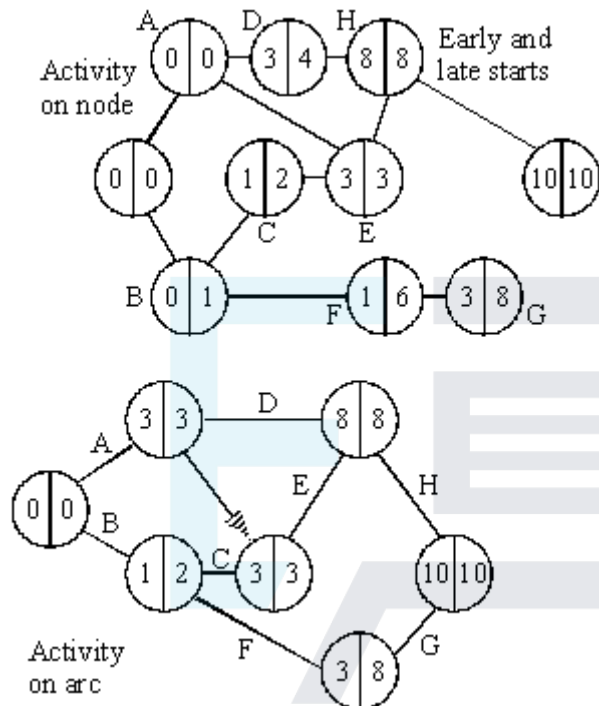
B1

1

[17]

Q35.

(a) + (b)



time:- 10

critical activities:- A, E, H

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M1

Sca

A1

single start and finish

A3

(- 1 each error)

M1

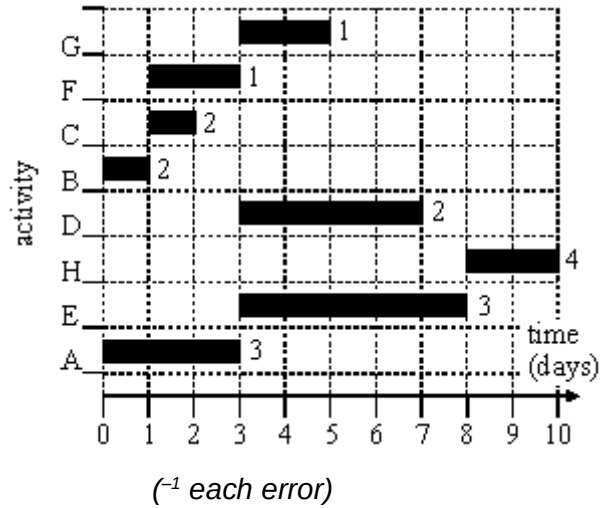
forward pass

A1 M1

backward pass

A1 B1 A1

(c)



M1 A2

- (d) F – 2; G – 8; D – 4
B2 (⁻¹)

[16]

Examiner reports

Q1.

Nearly all students correctly completed the precedence table. Just under two thirds of students completed the activity network with no errors, with the most common error being simple numerical slips that were then carried forward through the network. Some students had values for the latest finish time lower than the earliest start time.

Many students correctly identified the first activity on the critical path and explained why this activity could not be delayed. A minority of students did not find the time at which the making of the meal would need to commence, instead just stating the minimum completion time of their activity network.

Part (c) was answered well, with the majority of students scoring 1 mark on (c)(i) and at least 1 mark on (c)(ii). The most common mistake was for students not to relate the minimum completion time of the second project to the time that the second project should start.

Q4.

This question proved to be a good source of marks for all students, with many scoring full marks. There were problems in part (e) where some students left 'holes' with their blocks and others had the correct histogram profile, but failed to indicate the activities linked to each block.

Q5.

The first three parts of this question proved to be a good source of marks for all students with many scoring full marks. There were problems in part (d), where students failed to appreciate the effect of BOTH H and K overrunning and the implications to the critical path.

Q6.

Part (a) Almost every candidate found the correct values of the constants x , y and z .

Part (b) The two critical paths were found correctly by most candidates, but several candidates wrote down extra paths that were not critical paths.

Part (c) Most candidates identified the correct activity with the largest float and calculated the float correctly.

Part (d) The histogram was usually constructed correctly but a few candidates left "holes" in their blocks and others made slips having certain blocks with an incorrect height or width.

It was necessary to indicate clearly which activities were taking place at any given time; those who had the correct histogram profile, but without the activities indicated, did not score full marks.

Part (e) The resource levelling caused difficulties for many, with only the best candidates finding the correct minimum extra time required and the new start time for activity J .

Q7.

- (a) Almost every student calculated the earliest start times and latest finish times correctly.
- (b) At least one critical path was found correctly by almost everyone and most students found the two critical paths.
- (c) Greater care should be taken when drawing diagrams of this type, preferably **using a ruler**. The most common error in the cascade diagram was the omission of slack for the non-critical activities; others tried to show activities starting as late as possible. Some tried to insert activity J as two sections, so occasionally the diagram resembled a histogram.
- (d) Those students who made errors in their earliest start times and latest finish times for *J* were given credit for writing down an inequality based on their values. Quite a large number of students did not realise the need to use the difference between the latest finish time and earliest start time for *J* only and so had a totally incorrect inequality.

Q8.

In part (a) almost every candidate calculated the earliest start times and latest finish times correctly.

In part (b)(i) the single critical path was again found correctly by almost everyone.

In (b)(ii) a few candidates did not know how to calculate the float for activity *D*, but the majority were able to do so by finding $13 - 2 - 7$.

In part (c) the histogram was usually constructed correctly but a few candidates left gaps or “cantilevers” in their blocks and others made slips having certain blocks with an incorrect height or width. It was pleasing to see most candidates indicating clearly which activities were taking place at any given time.

In (d) the resource levelling caused a few problems; some could not deal correctly with the activity *F* which needed to be fitted in before *G* started. A number of alternatives were possible with the events following *G*, but the common mistake was trying to start *K* before *H* had finished. Those with a correct second histogram had no trouble in stating the correct minimum extra time required.

Q9.

- (a) Almost every candidate calculated the earliest start times and latest finish times correctly.
- (b) At least one critical path was found correctly by almost everyone and most candidates found the two critical paths. A surprising number neglected to state the minimum time for completion; it was not enough to simply have this value in the final box on the diagram.
- (c) Those candidates who made errors in their earliest start times and latest finish times were given credit for identifying their activity with the greatest float time and the majority seemed aware of how to calculate its value.
- (d) Greater care should be taken when drawing diagrams of this type, preferably using a ruler. The most common error in the cascade diagram was the omission of slack for the non-critical activities. Despite the bold printing of the word **late**, many showed

B, E, G and *J* starting as early as possible. Some tried to fit activity *G* in two sections, with one of the two blocks slotted between activities *I* and *J*, so the diagram resembled a histogram.

Q10.

Almost all candidates calculated the earliest start times and latest finish times in part (a) correctly.

In part (b), the float for activity *G* was calculated correctly by more than half the candidates and this was encouraging to see.

Most candidates found the two critical paths in part (c), but a surprising number omitted to state the minimum time for completion.

The histogram in part (d) was not always constructed correctly, with quite a few candidates leaving gaps in their blocks. It was pleasing to see most candidates indicating clearly which activities were taking place at any given time. Some candidates ignored the request to show the activities starting as **late** as possible, and others were obviously confusing their diagram with a Gantt chart and tried to indicate slack by extra dotted lines.

Q11.

This proved to be a very high scoring opening question for most candidates.

Part (a) The earliest start times were usually correct, but the latest finish times for some activities were sometimes calculated incorrectly.

Part (b) Most candidates found at least one correct critical path, but sometimes spoiled their answers by adding a third path that was not a critical path. Some candidates listed all the critical activities instead of giving the critical paths and scored no marks for this.

Part (c) Most cascade diagrams indicated the critical activities correctly. Quite a lot of candidates failed to show the correct slack for activities *C, D, G, I* or *J*. Others had slack overlapping another activity and this was not acceptable.

Part (d) The last part of the question was answered well by most candidates, although some failed to mention the new earliest starting **day** for either *K* or *L*.

Q12.

This proved to be a very good opening question for all candidates. The earliest start times were usually correct but the latest finish times for *D, B* and *A* were sometimes calculated incorrectly. Most candidates found the correct critical path, but quite a few forgot to state the minimum completion time. Those with a correct activity diagram were usually able to find the new start times for *H* and *I* after the delay. A few thought that the minimum delay was 1 day and others stated that the delay was 24 days, when in fact this was the new completion time.

Q13.

The network diagram was usually correct and almost all candidates calculated the earliest start times and latest finish times correctly and hence found the minimum completion time and critical path. Many candidates did not produce a correct resource histogram. The section between 10 and 12 days caused the biggest problem with many diagrams not

having complete vertical blocks. It was good to see most candidates heeding the advice of previous reports in labelling their diagrams so that it was possible to see which activities were taking place on a particular day.

Q14.

This proved to be a very good opening question for all candidates. The earliest start times and latest finish times were usually correct. Some candidates only gave one critical path instead of two. Quite a large number thought that *ADGJK* was also a critical path and lost a mark. Despite showing a value of 22 on their insert, several candidates forgot to state the minimum time for completion. The cascade diagram caused few problems. Some candidates used blocks with critical activities linked together and other activities drawn in blocks above; others used a sequence of horizontal lines with the vertical axis labelled from *A* to *K*. The main reason for losing marks in part (c) was the ignoring of the slack or float. The explanation in part (c) was not always adequate: it was necessary to explain that *F* needed to start after 12 days at the earliest, with *I* and *K* being delayed by one day, and hence that the new minimum completion time was 23 days.

Q15.

This was meant to be a confidence boosting first question and it proved to be the case. The network diagram was usually correct and almost all candidates calculated the earliest start times and latest finish times correctly. The minimum completion time was usually correct also. However, even some very able candidates failed to realise that there were three critical paths.

Q16.

The precedence table was usually correct and the earliest start times and latest finish times were found correctly by most candidates. Quite a few gave the latest finish time for activity *F* as 10 rather than 11, and consequently had values of 5 and 2 at *C* and *A* respectively; this was treated as a single error. Those with the correct values on the activity diagram were able to find the correct critical path and were able to conclude that activity *E* had the greatest float time. This proved to be a very good opening question for all candidates.

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Q17.

Parts (a),(b),(c),(d) and (e) were very well done by practically all the candidates. The topic of critical path analysis seems to be well understood by most candidates. Problems started to creep in when answering part (f) where sometimes the resource histogram was not a histogram at all; being more like a random collection of rectangles piled on top of each other, rather like Stonehenge. Candidates had been expected to indicate which activities were being represented on the histogram so as to make resource levelling easier to identify. Many candidates were successful in finding the minimum extra time, although the explanations as to why three extra days are sufficient were often incomplete.

Q18.

The network diagram and the critical path were answered very well by all candidates. A few were unable to identify float times for the non-critical activities. The major problem in the Gantt chart was a failure to show slack time. When this was indicated it was usually shown after the various activities even though the question asked to show activities starting as **late** as possible. Some indication should have been made on the diagram to indicate that *G* needed to start after *F* and also after *C* but this was not penalised this

time.

Q19.

This was a high scoring question for practically everyone. The activity network in part (a) was usually done correctly and candidates seemed well drilled in critical path analysis. Similarly, parts (b), (c) and (d) were answered correctly by even the weakest candidates. The explanation in part (d)(ii) was not always clear. Part (e) was meant to be a little more challenging, but only the weakest candidates failed to find the new earliest start time for activity *I* and the revised shortest completion time for the project.

Q20.

This question proved to be a good source of marks for all candidates. A significant number of candidates failed to answer parts (d) and (e) correctly. Correct solutions were sometimes obtained by candidates amending their figure by including the overruns and their implications.

Q21.

This question proved to be a good source of marks for all candidates, although a significant number failed to score full marks in part (d) due to the inaccuracies of their diagram or through drawing a resource histogram. Part (e) discriminated well between candidates. This was the first time that candidates were provided with a stage/state table for answering a dynamic programming question. The question gave candidates the opportunity to use a network diagram and still score full marks. A significant number of weaker candidates thought that, as there were only six possible orders, a complete enumeration was an acceptable method. This was not the case. About half of the candidates worked backwards through the network and about half worked forwards. Both methods were equally successful.

Q22.

This question was well answered by the majority of candidates. Marks were lost in part (d), where some candidates drew a Gantt chart rather than a resource histogram. Nearly all candidates used graph paper for their attempt at the resource histogram. Marks were lost in part (e)(ii) by candidates not stating clearly which worker should be allocated which task. The mark could not be obtained by quoting the critical path found in part (b), as the critical part (b) to part (e)(i).

Q23.

This question proved to be a good source of marks for all candidates, although a significant number of candidates failed to score full marks in part (c), due to the inaccuracies of their diagram or through drawing a resource histogram instead of a Gantt diagram.

Q24.

This question was well answered by candidates. Marks were lost in the final part of the question where candidates did not show the float times, where appropriate. Nearly all candidates used their answer book for their Gantt diagram and full marks were regularly obtained, provided the correct timings were clearly illustrated.

Q25.

This question was well answered, with the majority of candidates scoring at least 10 out of 13 marks. Whereas parts (a), (b) and (c) were nearly always correct, candidates did make mistakes in part (d). The question stated that activities were to start as **late** as possible. This was the first time that this had been asked. A number of candidates made a mistake with activity *E*.

Also, for the first time, some candidates failed to show accurately the start and finish times for their Gantt diagram. If candidates draw their answer in their answer booklet, they must clearly indicate start and finish times for each activity. Candidates who used graph paper did not lose accuracy marks.

Q26.

This question was well answered by the majority of the candidates, with the exception of part (e). The vast majority of candidates drew an 'activity on node' diagram and this proved to be very successful. Part (e) required candidates to draw a resource histogram for the project but unfortunately a significant number of candidates merely drew a Gantt diagram. Candidates should be encouraged on all such diagrams to use graph paper and to make sure their work is well presented.

Q27.

The majority of candidates scored full marks on parts (a) and (b), but part (c) discriminated well amongst the candidates. Although candidates could successfully find the new minimum completion time, they did not easily apply this time to implications for critical/non-critical activities.

Q28.

Parts (a), (b) and (c) were answered well by many candidates but part (d) proved to be the hardest question on the paper. There were some mistakes made on the backward pass to find the latest finish time for the activities. The majority of candidates used graph paper to draw their Gantt diagram, a practice that should be encouraged. In part (d) those candidates who looked at the original network realised that with only two workers there was a problem with *B, C, D* and *I, J, K* and found the minimum extra time that two workers would need to complete the three activities and arrived at the correct solution.

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Q29.

The question involved a routine application of critical path analysis. The candidates were well prepared and scored nearly all the available marks in the first five parts. It was pleasing to note that candidates worked their critical path on their network diagram, with the benefit of saving time for the later, more searching, parts of the question. Although part (f) was challenging, there were a number of correct solutions with candidates able to relate the new information back to the critical path together with its implications.

Q30.

This was a standard critical path analysis question in which candidates were expected to use their diagram in part (a) to help them with the later parts of the question. It seems that candidates have been made aware of previous comments in examiners' reports and used their network as a working diagram. Part (c) proved to be a good discriminator with only the best candidates being able to justify their answers fully.

Q31.

The question was a routine application of critical path analysis for which the candidates

were well prepared and scored nearly all available marks in the first four parts. It was pleasing to note that candidates worked their critical path on their network diagram with the benefit of saving time for the later, more thoughtful parts of the question. Part (g) proved to be too challenging for the candidates.

Q32.

Candidates were able to score the majority of marks on parts (a) to (d), but part (e) proved to be challenging. Candidates needed to consider whether the speeding up of an activity would speed up the completion of the whole project and, after an activity has been speeded up, what the implications on other activities would be.

Q33.

The question was a routine application of critical path analysis for which the candidate was well prepared and scored nearly all the available marks.

Q34.

This was a standard critical path analysis question where candidates were expected to use their diagram in part (a) to help them with the later parts of the question. Unfortunately too many candidates chose to compute each critical time using the full algorithmic method rather than using their activity network. The numbers used in a critical path question will normally be easy enough to allow candidates to use their network as a working diagram.

Q35.

This question was generally very well answered, with many candidates obtaining at least 14 of the available 16 marks. Almost everyone obtained the correct activity network, and almost all of the forward passes were correct. There were some slips made in the backward pass, particularly for activity B. Most obtained the correct minimum duration and list of critical activities.

In part (c) the majority of candidates completed the cascade chart correctly, although a few showed all of the activities starting at time zero. Far fewer could obtain the new scheduled start times in part (d).

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